Introduction to Three Dimensional Geometry

Class 11 Mathematics • Practice Exercise with Solutions

Topic: Coordinates, Distance Formula & Section Formula | Difficulty: Easy to Moderate

Key Formulas Quick Reference

- **Distance Formula:** $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$
- **Internal Section Formula:** $P(x, y, z) = ((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n), (mz_2 + nz_1)/(m+n))$
- **External Section Formula:** $P(x, y, z) = ((mx_2 - nx_1)/(m-n), (my_2 - ny_1)/(m-n), (mz_2 - nz_1)/(m-n))$
- **Centroid of Triangle:** $G(x, y, z) = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3, (z_1 + z_2 + z_3)/3)$

SECTION A: EASY PRACTICE PROBLEMS

Problem 1: Octant Identification

EASY

Determine the octants in which the following points lie: **(i)** $A(4, -2, 3)$ **(ii)** $B(-2, -4, -5)$ **(iii)** $C(-3, 1, -2)$

SOLUTION:

The sign of coordinates (x, y, z) determines the octant:

- **(i)** $A(4, -2, 3)$: $x > 0, y < 0, z > 0$ → Lies in **Octant IV**.
- **(ii)** $B(-2, -4, -5)$: $x < 0, y < 0, z < 0$ → Lies in **Octant VII**.
- **(iii)** $C(-3, 1, -2)$: $x < 0, y > 0, z < 0$ → Lies in **Octant VI**.

Problem 2: Distance Between Two Points

EASY

Find the distance between the points $P(1, -3, 4)$ and $Q(-4, 1, 2)$.

SOLUTION:

Using the 3D distance formula $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$:

$PQ = \sqrt{[(-4 - 1)^2 + (1 - (-3))^2 + (2 - 4)^2]}$

$PQ = \sqrt{[(-5)^2 + (4)^2 + (-2)^2]} = \sqrt{[25 + 16 + 4]} = \sqrt{45} = 3\sqrt{5}$

Answer: $3\sqrt{5}$ units

Problem 3: Coordinates on Coordinate Planes & Axes

EASY

(a) What are the coordinates of a point lying on the Y-axis?

(b) What is the perpendicular distance of the point $P(3, 4, 5)$ from the XY-plane?

SOLUTION:

(a) Any point lying on the Y-axis has both its x and z coordinates equal to 0. Hence, it is of the form $(0, y, 0)$.

(b) The perpendicular distance of any point (x, y, z) from the XY-plane is given by $|z|$.

For $P(3, 4, 5)$, distance from XY-plane = $|5| = 5$ units.

Problem 4: Midpoint Formula

EASY

Find the coordinates of the midpoint of the line segment joining the points $A(2, 3, 4)$ and $B(4, 1, -2)$.

SOLUTION:

The midpoint M is given by $((x_1 + x_2)/2, (y_1 + y_2)/2, (z_1 + z_2)/2)$.

$x = (2 + 4)/2 = 3$

$y = (3 + 1)/2 = 2$

$z = (4 + (-2))/2 = 1$

> **Midpoint:** $M(3, 2, 1)$

Problem 5: Direct Application of Section Formula

EASY

Find the coordinates of the point which divides the line segment joining $(1, -2, 3)$ and $(3, 4, -5)$ internally in the ratio $2 : 3$.

SOLUTION:

Let $m = 2$, $n = 3$, $(x_1, y_1, z_1) = (1, -2, 3)$, and $(x_2, y_2, z_2) = (3, 4, -5)$.

$x = (2(3) + 3(1)) / (2 + 3) = (6 + 3)/5 = 9/5$

$y = (2(4) + 3(-2)) / (2 + 3) = (8 - 6)/5 = 2/5$

$z = (2(-5) + 3(3)) / (2 + 3) = (-10 + 9)/5 = -1/5$

> **Required Point:** $(9/5, 2/5, -1/5)$

SECTION B: MODERATE PRACTICE PROBLEMS

Problem 6: Collinearity Verification

MODERATE

Show that the points $A(-2, 3, 5)$, $B(1, 2, 3)$, and $C(7, 0, -1)$ are collinear.

SOLUTION:

Calculate distances between each pair of points:

- $AB = \sqrt{[(1 - (-2))^2 + (2 - 3)^2 + (3 - 5)^2]} = \sqrt{[3^2 + (-1)^2 + (-2)^2]} = \sqrt{[9 + 1 + 4]} = \sqrt{14}$
- $BC = \sqrt{[(7 - 1)^2 + (0 - 2)^2 + (-1 - 3)^2]} = \sqrt{[6^2 + (-2)^2 + (-4)^2]} = \sqrt{[36 + 4 + 16]} = \sqrt{56} = 2\sqrt{14}$
- $AC = \sqrt{[(7 - (-2))^2 + (0 - 3)^2 + (-1 - 5)^2]} = \sqrt{[9^2 + (-3)^2 + (-6)^2]} = \sqrt{[81 + 9 + 36]} = \sqrt{126} = 3\sqrt{14}$

Notice that $AB + BC = \sqrt{14} + 2\sqrt{14} = 3\sqrt{14} = AC$.

Since $AB + BC = AC$, points A, B, C are collinear.

Problem 7: Ratio Division by a Coordinate Plane

MODERATE

Find the ratio in which the YZ-plane divides the line segment joining the points $A(-2, 4, 7)$ and $B(3, -5, 8)$. Also, find the point of intersection.

SOLUTION:

Let the YZ-plane divide AB in the ratio $k : 1$ at point P.

By section formula, $x = (3k - 2)/(k + 1)$.

Since point P lies on the YZ-plane, its x-coordinate is 0:

$(3k - 2)/(k + 1) = 0 \Rightarrow 3k - 2 = 0 \Rightarrow k = 2/3$

Thus, the ratio is $2 : 3$ internally.

Now substitute $k = 2/3$ to find y and z:

$y = [2/3(-5) + 1(4)] / (2/3 + 1) = (-10/3 + 4) / (5/3) = (2/3) / (5/3) = 2/5$

$z = [2/3(8) + 1(7)] / (2/3 + 1) = (16/3 + 7) / (5/3) = (37/3) / (5/3) = 37/5$

Ratio: $2 : 3$ | Point of intersection: $(0, 2/5, 37/5)$

Problem 8: Isosceles Right-Angled Triangle Verification

MODERATE

Show that the vertices $A(0, 7, 10)$, $B(-1, 6, 6)$, and $C(-4, 9, 6)$ form an isosceles right-angled triangle.

SOLUTION:

Compute the lengths of all three sides:

• $AB^2 = (-1 - 0)^2 + (6 - 7)^2 + (6 - 10)^2 = 1 + 1 + 16 = 18$

• $BC^2 = (-4 - (-1))^2 + (9 - 6)^2 + (6 - 6)^2 = (-3)^2 + 3^2 + 0 = 9 + 9 = 18$

• $AC^2 = (-4 - 0)^2 + (9 - 7)^2 + (6 - 10)^2 = 16 + 4 + 16 = 36$

1. Since $AB^2 = BC^2 = 18$, $AB = BC = 3\sqrt{2}$ (Isosceles).

2. $AB^2 + BC^2 = 18 + 18 = 36 = AC^2$ (Pythagoras Theorem satisfied).

Hence, ΔABC is an isosceles right-angled triangle at B.

Problem 9: Centroid and Missing Vertex

MODERATE

If the centroid of triangle ABC with vertices $A(1, a, 3)$, $B(b, -2, 5)$, and $C(2, 1, c)$ is at $G(0, 0, 0)$, find the values of a, b, and c.

SOLUTION:

The centroid coordinates $G(x, y, z)$ are given by:

$((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3, (z_1 + z_2 + z_3)/3) = (0, 0, 0)$

• x-coordinate: $(1 + b + 2)/3 = 0 \Rightarrow b + 3 = 0 \Rightarrow b = -3$

• y-coordinate: $(a - 2 + 1)/3 = 0 \Rightarrow a - 1 = 0 \Rightarrow a = 1$

• z-coordinate: $(3 + 5 + c)/3 = 0 \Rightarrow c + 8 = 0 \Rightarrow c = -8$

Answer: $a = 1, b = -3, c = -8$

Problem 10: Locus of a Moving Point

MODERATE

Find the equation of the set of points P such that the sum of its distances from $A(4, 0, 0)$ and $B(-4, 0, 0)$ is equal to 10.

SOLUTION:

Let $P(x, y, z)$ be any point on the locus. Given condition: $PA + PB = 10$.

$\sqrt{[(x - 4)^2 + y^2 + z^2]} + \sqrt{[(x + 4)^2 + y^2 + z^2]} = 10$

Shift one radical to RHS: $\sqrt{[(x - 4)^2 + y^2 + z^2]} = 10 - \sqrt{[(x + 4)^2 + y^2 + z^2]}$

Squaring both sides:

$(x - 4)^2 + y^2 + z^2 = 100 - 20\sqrt{[(x + 4)^2 + y^2 + z^2]} + (x + 4)^2 + y^2 + z^2$

Expand and simplify terms:

$x^2 - 8x + 16 = 100 - 20\sqrt{[(x + 4)^2 + y^2 + z^2]} + x^2 + 8x + 16$

$-16x - 100 = -20\sqrt{[(x + 4)^2 + y^2 + z^2]} \Rightarrow 4x + 25 = 5\sqrt{[(x + 4)^2 + y^2 + z^2]}$

Squaring both sides again:

$(4x + 25)^2 = 25[(x + 4)^2 + y^2 + z^2]$

$16x^2 + 200x + 625 = 25(x^2 + 8x + 16 + y^2 + z^2)$

$16x^2 + 200x + 625 = 25x^2 + 200x + 400 + 25y^2 + 25z^2$

$9x^2 + 25y^2 + 25z^2 = 225$

Required Equation: $9x^2 + 25y^2 + 25z^2 = 225$